

SIMATS ENGINEERING
 Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	Slot	B
Course Title	Cloud Computing and Big Data Analytics	Assessment	CO4 AT1
Assessment Title	Big Data Analytics Class Test	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	Total Marks	30

Student Details

Student Name	Blessy S	Register Number	192421053
Class / Section	Btech it	Slot	B
Capstone Title	Cloud security	Submission Date	19.08.26
GitHub Repository Link	https://github.com/192421053simatscmd/Cloud-computing-	Google Classroom Status	Turned in

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

Criteria	Marks	Evaluation Focus	Student Check
Concept Accuracy	9	Correct technical meaning, terminology and examples. Have I used correct big data terms?	
Coverage	6	All required parts of the question are answered. Have I answered every part?	

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.

- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I ___Blessy S(192421053)___(Name / Register Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **blessy**

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Big Data refers to extremely large and complex datasets whose **size, speed, variety, or complexity exceeds the capabilities of conventional data processing tools**. In cloud security, huge amounts of data are continuously generated from sources such as login records, authentication events, firewall logs, network traffic, API requests, access logs, system events, and security alerts. Traditional tools such as spreadsheets or single-server databases may struggle to handle this data because of **limited storage capacity, slow query performance, high numbers of simultaneous users, fixed schemas, and limited scalability**. For example, a cloud platform used by a large organization may generate millions of login attempts and network events every day. Storing and analyzing all these events on a single server can result in slow processing and make it difficult to identify security threats in real time. Big data technologies solve this problem by using **distributed storage and parallel processing**, where large datasets are divided and stored across multiple machines and processed simultaneously. This allows security teams to analyze huge volumes of cloud activity, detect unusual behavior, identify possible cyberattacks, and generate alerts more quickly. Therefore, Big Data is essential in cloud security because it provides the scalability and processing capability required to monitor large cloud environments and protect sensitive data and resources.

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

The **5Vs of Big Data** are **Volume, Velocity, Variety, Veracity, and Value**. **Volume** refers to the huge amount of data generated and stored by organizations. **Velocity** refers to the speed at which data is created, collected, and processed. **Variety** describes the different forms of data, such as structured data, JSON files, images, videos, logs, and text. **Veracity** refers to the accuracy, reliability, and quality of the data, because incorrect or incomplete data can lead to wrong decisions. **Value** refers to the useful information and benefits that can be obtained by analyzing the data. For example, in **cloud security**, a cloud platform may generate millions of login records, firewall logs, API requests, and network events. This represents **Volume**, because a large amount of security data is produced and stored. **Velocity** can be seen when login attempts and network activities are continuously generated and need to be analyzed in real time to detect suspicious activity. **Variety** affects the system because security data may come in different formats, including logs, user activity records, IP addresses, alerts, and API responses. **Veracity** is important because inaccurate or incomplete security logs could cause false alarms or allow threats to go undetected. Finally, **Value** is obtained when analytics converts this data into useful results such as threat detection, security alerts, risk identification, and improved protection. Therefore, the 5Vs influence how big data systems are designed for storage, processing, security, and analytics.

Question 3 (3 Marks)

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required

Structured, semi-structured, and unstructured data are three common types of data used in cloud-based applications. Structured data is highly organized and follows a fixed schema, usually in the form of rows and columns in a database. For example, in a cloudbacked healthcare application, patient details such as patient ID, name, age, disease, and appointment date can be stored in a structured table. Semi-structured data does not follow a fixed tabular format but contains keys, tags, or labels that help organize the information. Examples include JSON and XML files, API responses, and system log files. In the same healthcare application, patient information received from an API in JSON format can be considered semi-structured data. Unstructured data does not have a predefined format or schema and is usually more difficult to organize and analyze. Examples include medical images, videos, audio recordings, PDF reports, emails, and social media text. For example, X-ray images and scanned medical reports uploaded to a cloud system are unstructured data. Therefore, structured data is organized in a fixed format, semi-structured data has some organization using keys or tags, and unstructured data has no fixed stru

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.

3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Distributed file systems play a major role in big data processing because modern applications generate huge amounts of data that cannot be efficiently stored or processed on a single computer. Instead of keeping the entire dataset on one machine, a distributed file system divides large files into smaller blocks and distributes those blocks across multiple computers or storage nodes in a network. This approach allows organizations to store and manage very large datasets efficiently. One of the main advantages is scalability, because additional machines can be added to the system when the amount of data increases. Distributed file systems also provide fault tolerance by maintaining multiple copies, or replicas, of important data blocks. If one machine fails, another machine containing a copy of the data can continue to provide access, reducing the risk of data loss. Another important benefit is parallel access and processing. Since different parts of a large dataset are stored on different machines, multiple machines can read and process those parts simultaneously, which significantly reduces the time required to analyze large datasets. For example, Hadoop Distributed File System (HDFS) is a commonly used distributed file system in big data environments. HDFS divides large files into blocks and distributes them across different nodes while maintaining replicas for reliability. It is designed to work with big data processing frameworks such as MapReduce and Apache Spark. These frameworks can process the data blocks in parallel and combine the results to produce the final output. Distributed file systems are mainly used for large-scale file storage and are different from relational databases, which organize data into structured tables with predefined schemas. Cloud object storage services can also provide similar scalable storage capabilities for big data applications. Overall, distributed file systems provide the foundation for big data processing by offering scalable storage, fault tolerance, parallel data access, and efficient support for batch processing. Without distributed storage, processing extremely large datasets across multiple machines would be much more difficult and time-consuming.

Question 5 (3 Marks)

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Big data platforms handle huge volumes of information collected from different sources, which makes security and privacy more difficult to manage. Three major security concerns are unauthorized access, data leakage, and privacy violations. Unauthorized access occurs when users who do not have proper permission gain access to sensitive data. This can happen because of weak passwords, incorrect permissions, or poor access controls. It can be controlled using role-based access control (RBAC), strong authentication, and proper user permissions. Data leakage is another major concern because large datasets may contain confidential customer, financial, business, or personal information. If the data is exposed through a security attack, insecure storage, or an unsecured API, it can be misused or stolen. Encryption should be used for data both during transmission and while it is stored, along with regular monitoring and logging. Privacy violation can occur when big data combines information from multiple sources and makes it possible to identify individuals or reveal sensitive personal information. Even if individual datasets appear harmless, combining them may expose private details. This risk can be reduced through data anonymization, masking, access restrictions, and proper data governance. Therefore, big data security requires strong access control, encryption, privacy protection, monitoring, and clear security policies because multiple data sources increase the complexity of protecting information.

Question 6 (3 Marks)

What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
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2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Data appliances and integration tools are important components of big data workflows because they help organizations store, process, move, and analyze large amounts of data efficiently. A data appliance is a specialized hardware and software platform designed and optimized for particular data storage, processing, or analytics workloads. It provides the required computing, storage, and processing resources in an integrated system, making it easier to manage large datasets and perform analytics. Integration tools, on the other hand, are software tools used to move, transform, and synchronize data between different sources, storage systems, and analytics platforms. Examples include ETL and ELT tools, API connectors, message queues, data ingestion tools, and workflow schedulers. In a typical big data workflow, data is first collected from different sources such as databases, applications, sensors, websites, or APIs. The data is then ingested using integration tools and cleaned or transformed into a suitable format. After transformation, the data is stored in a data lake, distributed file system, data warehouse, or cloud storage platform. Big data processing tools can then analyze the stored data, and the results can be presented through dashboards, reports, or analytics applications to support decision-making. Therefore, data appliances provide optimized infrastructure for handling data workloads, while integration tools connect different systems and ensure that data moves smoothly through the complete workflow from source → ingestion → transformation → storage → processing → dashboard or decision.

Question 7 (3 Marks)

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Organizations face several challenges when implementing big data analytics. Two major challenges are data quality and integration and skill and infrastructure requirements. Data quality and integration is a major problem because organizations collect data from many different sources such as databases, websites, mobile applications, sensors, social media, and cloud services. This data may be incomplete, duplicated, inconsistent, or stored in different formats. If poor-quality data is used for analytics, the results may be inaccurate and can lead to wrong business decisions. Integrating data from different systems can also be complex and time-consuming. Organizations can reduce this problem by implementing proper data governance, data cleaning, validation, standardization, and integration processes before performing analytics. Another major challenge is the lack of skilled professionals and the cost of infrastructure. Big data analytics requires employees with knowledge of data engineering, cloud computing, databases, analytics, and machine learning. Organizations may also need powerful storage and computing resources to process large datasets, which can increase infrastructure and maintenance costs. A lack of skilled employees can delay projects and reduce the quality of analysis, while high infrastructure costs can make large-scale implementation difficult. Organizations can address these challenges by providing employee training and hiring skilled professionals, while using cloud platforms that allow resources to be scaled according to demand instead of purchasing expensive hardware. Therefore, organizations should use proper data management, employee training, and scalable cloud infrastructure to successfully implement big data analytics and achieve better business results.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Cloud security is an important domain where big data analytics can be used to protect cloud-based systems, applications, and data from cyber threats. Cloud platforms generate large amounts of security data from different sources, such as login records, network traffic, firewall logs, access records, API activity, system logs, and user behavior. Big data analytics can collect and analyze this information to identify unusual patterns and potential security threats. For example, if a user normally logs in from one location but suddenly there are multiple login attempts from different locations, the system can analyze the activity and identify it as suspicious behavior. Analytics can also detect malware, unauthorized access, data breaches, abnormal network traffic, and possible insider threats. Real-time analysis can generate security alerts when unusual activity is detected, allowing administrators to respond quickly. Historical security data can also be analyzed to identify attack patterns and improve future security measures. The main benefits include faster threat detection, improved monitoring, reduced security risks, protection of sensitive cloud data, and quicker incident response. Therefore, big data analytics helps cloud security systems convert large volumes of security data into useful information for detecting threats, identifying risks, and protecting cloud resources.

Question 9 (3 Marks)

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Business drivers play an important role in influencing organizations to adopt big data solutions because they help businesses solve practical problems and achieve measurable benefits. One major driver is cost reduction, where organizations use big data analytics to identify unnecessary expenses, optimize resource usage, and improve operational efficiency. This creates a technical requirement for systems that can collect and analyze large amounts of operational data. Another important driver is better decision-making, where organizations analyze historical and real-time data to identify trends, predict future outcomes, and support management decisions. This requires data warehouses, cloud storage, analytics tools, and visualization platforms. Personalization is another business driver, especially for online services, where customer behavior,

preferences, and interactions are analyzed to provide personalized recommendations and improve user experience. Similarly, risk control and security encourage organizations to use big data for fraud detection, abnormal activity detection, and continuous monitoring. In cloud security, for example, large amounts of login records, network traffic, access logs, and API activity can be analyzed to identify suspicious behavior and potential cyberattacks. Speed and real-time analysis also influence adoption because organizations need quick alerts and responses to changing situations. Therefore, business drivers determine the type of big data technologies and analytics required by an organization, and adoption should ultimately be justified by measurable business value such as reduced costs, improved security, faster decisions, increased efficiency, or better customer experience.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

Type your answer here. Use short paragraphs, bullets, small tables or diagrams if required.

Big data analytics is important in decision making because it helps organizations convert large volumes of raw data into useful information, patterns, predictions, alerts, and recommendations. Instead of making decisions only based on assumptions or limited information, organizations can use data collected from different sources to make more accurate and evidence-based decisions. Big data analytics can support different levels of decision making. Descriptive analytics explains what has happened by analyzing historical data, while diagnostic analytics helps understand why something happened. Predictive analytics uses historical and current data to predict future events, and prescriptive analytics suggests suitable actions based on the predicted results. For example, in cloud security, an organization can analyze login records, network traffic, access logs, API activity, and system logs. Descriptive analytics can show the number of security incidents, diagnostic analytics can identify the possible reason for an incident, predictive analytics can identify users or systems that may be at risk, and prescriptive analytics can recommend actions such as blocking suspicious access or increasing security controls. This allows security teams to detect threats quickly and respond before serious damage occurs. Therefore, big data analytics

makes decision making faster, more accurate, evidence-based, and scalable, helping organizations reduce risks, improve performance, and respond effectively to changing situations.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done / Pending	done
All 10 questions answered.	Done / Pending	done
Examples are connected to cloud/capstone/domain context.	Done / Pending	done
Security/privacy points are included wherever relevant.	Done / Pending	done
Document exported as PDF if requested.	Done / Pending	done
GitHub link shared through Google Classroom.	Done / Pending	done